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c) 7.18 + 4.3
d) 6.5 + .0136

Solution:

a) Among the two, the least number of significant after decimal is one

$$\sqrt{4.5-3.31} = \sqrt{1.19} = 1.09$$

b) The number of minimum significant figures is 2

$$5.9\times10^5 - 2.3\times10^4$$
$$= 5.9\times10^5 - 0.23\times10^5$$
$$= 5.67\times10^5 = 5.7\times10^5$$

c) The lowest least number of significant digit after decimal is one

7.18 + 4.3 = 11.48 Rounding off we get 11.5

d) The lowest least number of significant digit after decimal is one

$$6.5 + .0136 = 6.5136 = 6.5$$

14. Arrive at Einstein's mass-energy relation by dimensional method ($E = mc^2$)

Solution:

Let us assume that the Energy E depends on mass m and velocity of light c.

$$E \alpha m^a c^b$$

$E = km^a c^b$ where K a constant

Dimensions of E = $[ML^2T^{-2}]$

Dimensions of m = [M]

Dimensions of c = $[LT^{-1}]$

Substituting the values in the above equation

$$[ML^2T^{-2}] = K\,[M]^a[LT^{-1}]^b$$

By equating the dimensions,

a = 1
b = 2
-b = -2
$E = k.mc^2$

The value of constant k = 1

$E = mc^2$ This is Einstein's mass energy relation

15. The velocity of a body is given by the equation $v = b/t + ct^2 + dt^3$. Find the dimensional formula for b.

Solution:

(b/t) should have the dimensions of velocity
b has the dimensions of (velocity x time)

$$[b] = [LT^{-1}][T] = [L] = [M^0L^1T^0]$$

16. The initial and final temperatures of a liquid in a container are observed to be 75.4 ± 0.5°C and 56.8 ± 0.2°C. Find the fall in the temperature of the liquid.

Solution:

$t_1 = (75.4 \pm 0.5)°C$
$t_2 = (56.8 \pm 0.2)°C$
Fall in temperature = (75.4 ± 0.5°C) – (56.8 ± 0.2°C)

$$t = (18.6 \pm 0.7)°C$$